

Investigating the Effects of Wing Angles on Downforce and Drag Generation in Formula Racecars

Introduction

Aerodynamic performance is crucial to success in Formula One. The front and rear wings are key aerodynamic components, providing **stability, handling,** and increased **cornering speed** by generating downforce (Syal, 2020). At speeds over 200 mph, F1 cars would take flight without enough downforce, causing deadly crashes that cost millions in damages. The front wing can generate up to 30% of the car's total downforce (Kachare, 2017). However, **increased downforce results in increased drag,** making optimization essential.

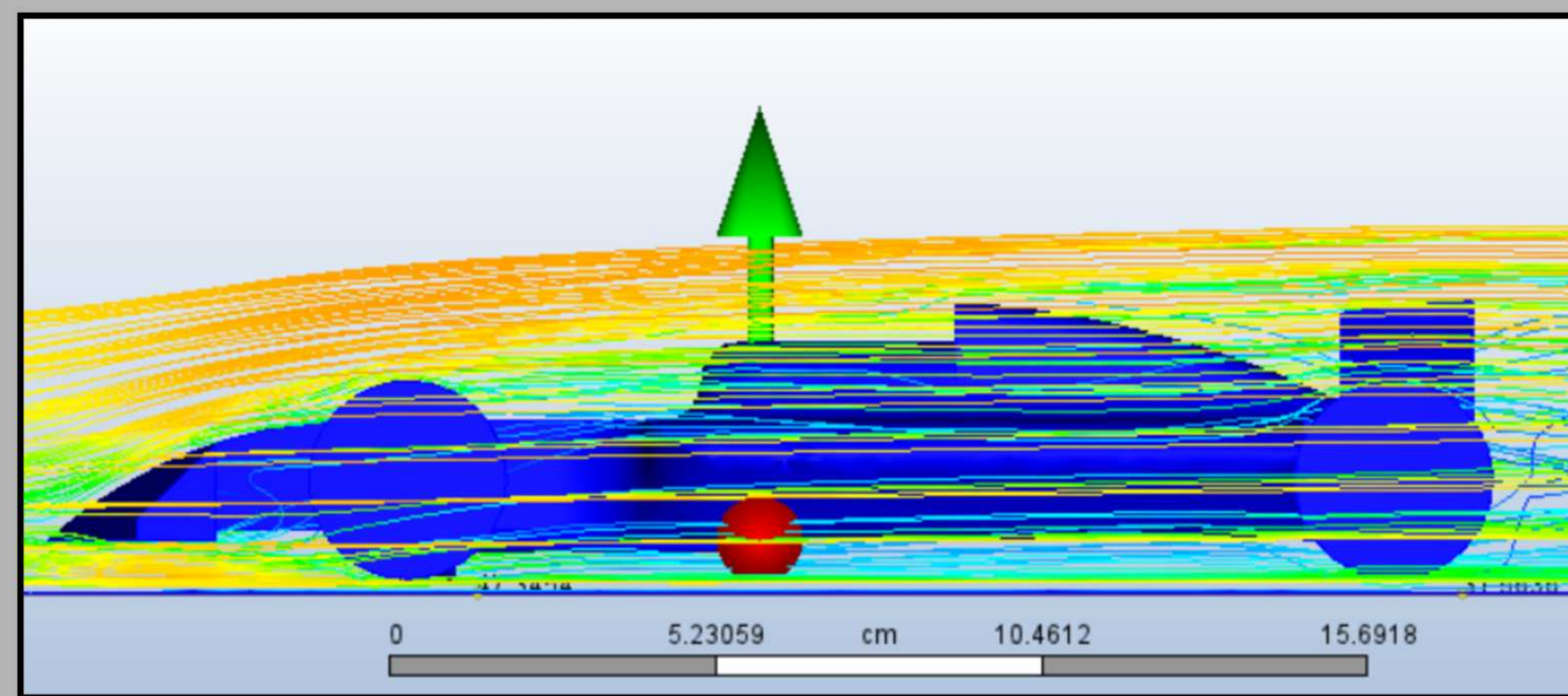


Figure 1. **Airflow traces** of sample car created in Autodesk CFD

Front wing **angle of attack** (AoA), the angle between the chord line of an airfoil and the direction of relative wind, directly affects downforce and drag (Syal, 2020).

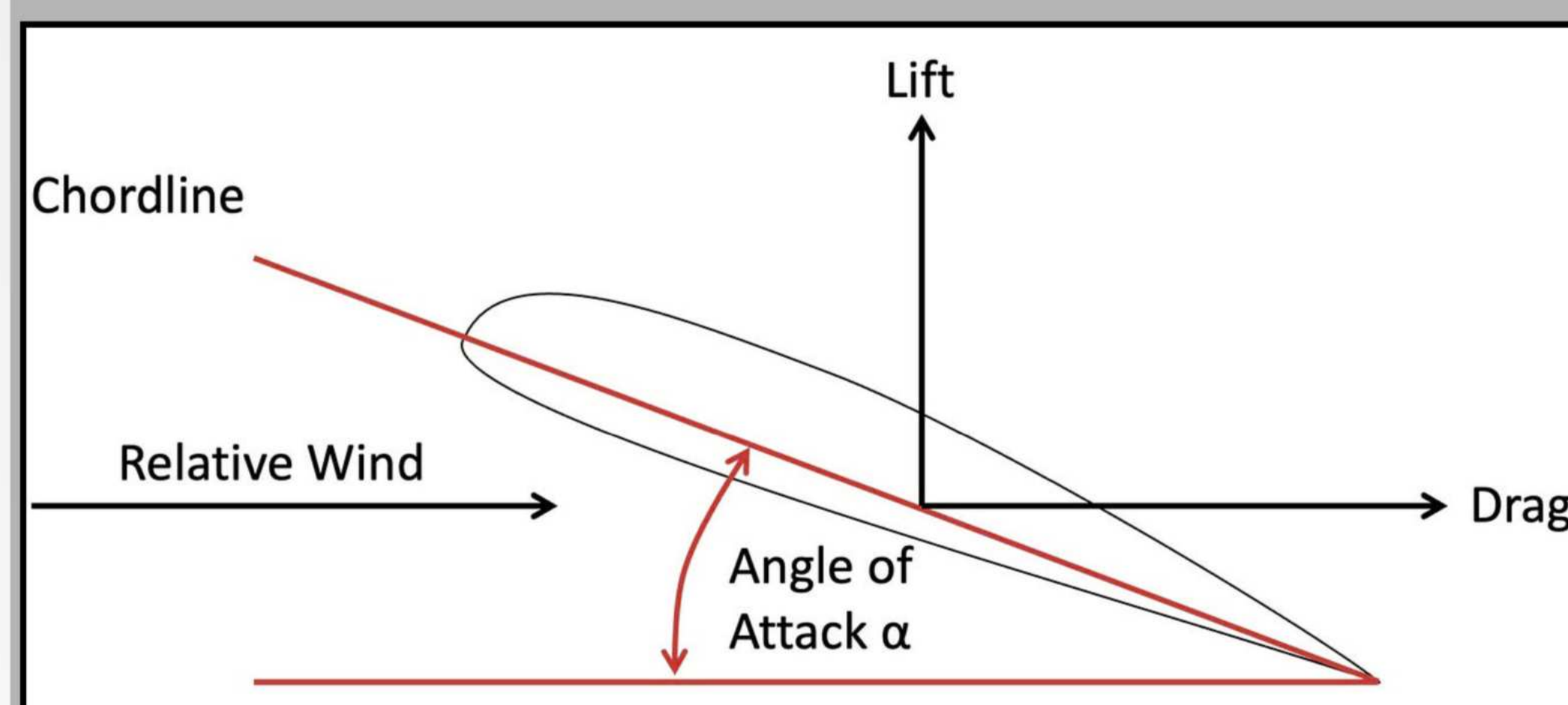


Figure 2. **Angle of Attack** definition
(<https://www.edn.com/aircraft-angle-of-attack-a0a-sensor-design/>)

Methods

1. Create simple **CAD** models of F1 cars with varied front wings in Autodesk Inventor
2. **3D print** each variation at 1:24 scale in BambuLab X1 Carbon with PLA filament
3. Run each variation through the Pitsco **Wind Tunnel**. Align the car's wheels with the front and rear downforce sensor plates, then use a small hook taped to the front of the car to connect to a drag force sensor. Increase the wind speed while collecting downforce and drag data.

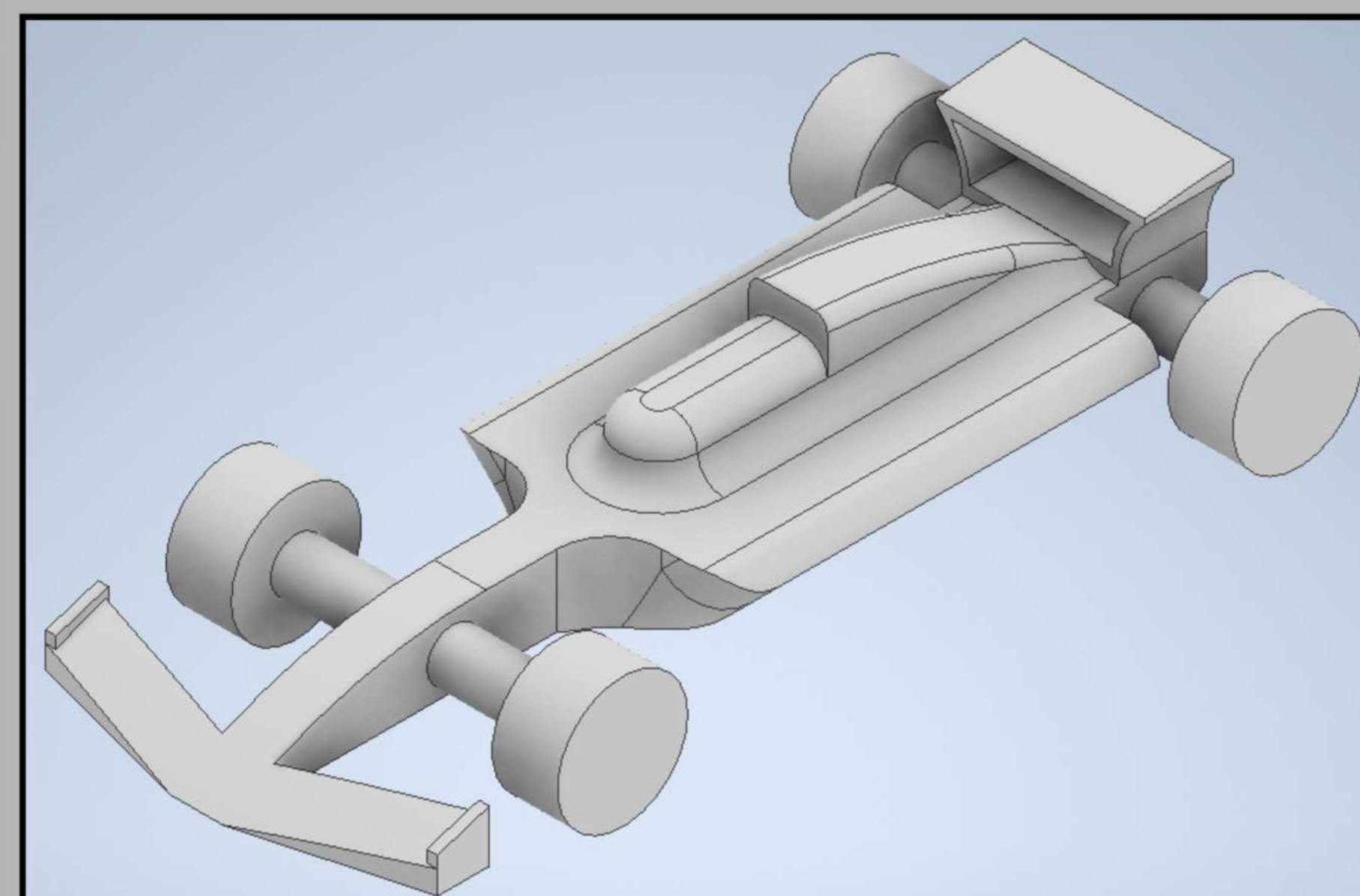


Figure 3. **CAD** model with 15° angle of attack front wing (Inventor Software)

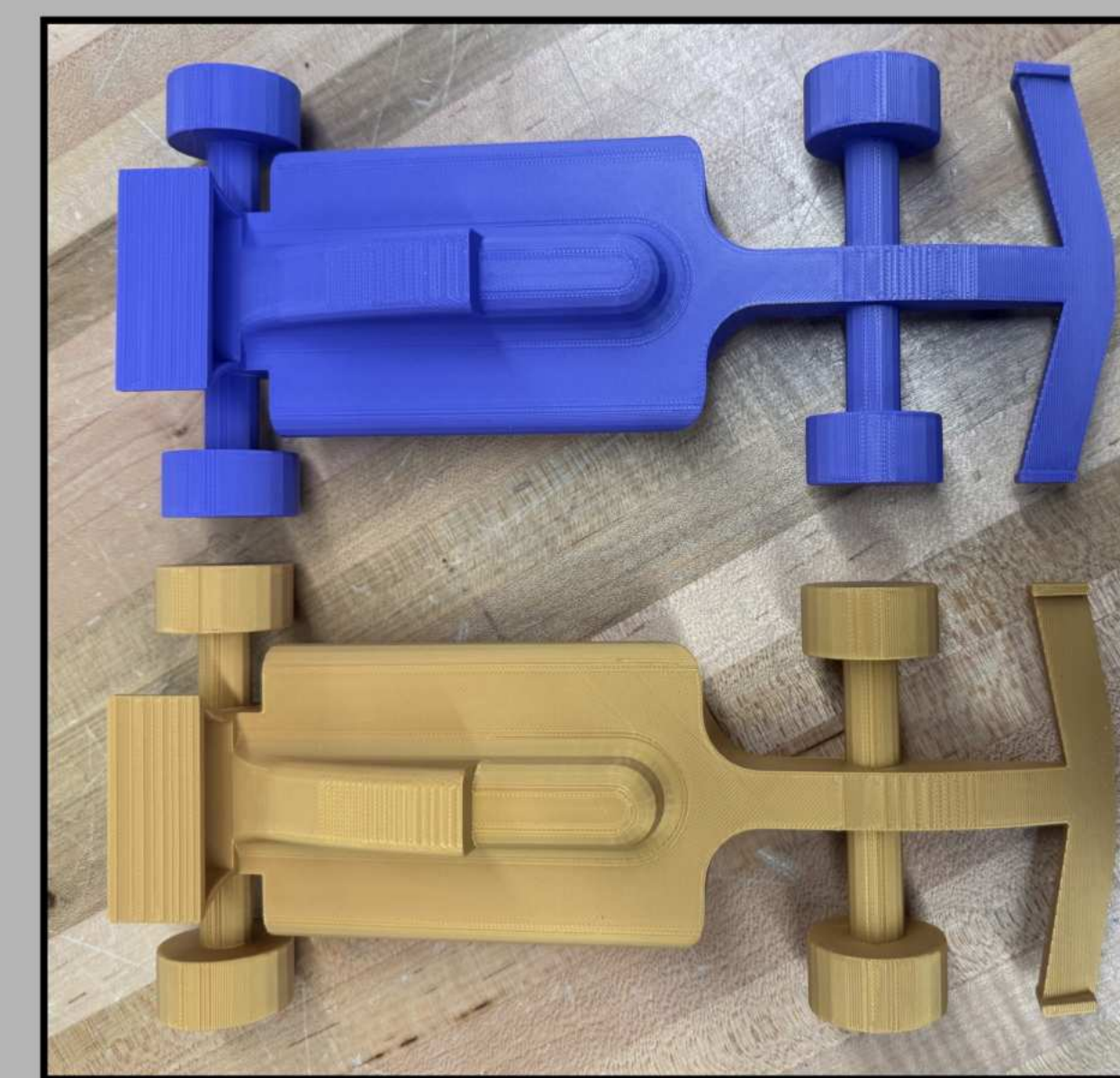


Figure 4. **3D printed** models with 15° AoA (Purple) and 10° AoA (Gold)



Figure 5. **Pitsco Wind Tunnel** used for testing

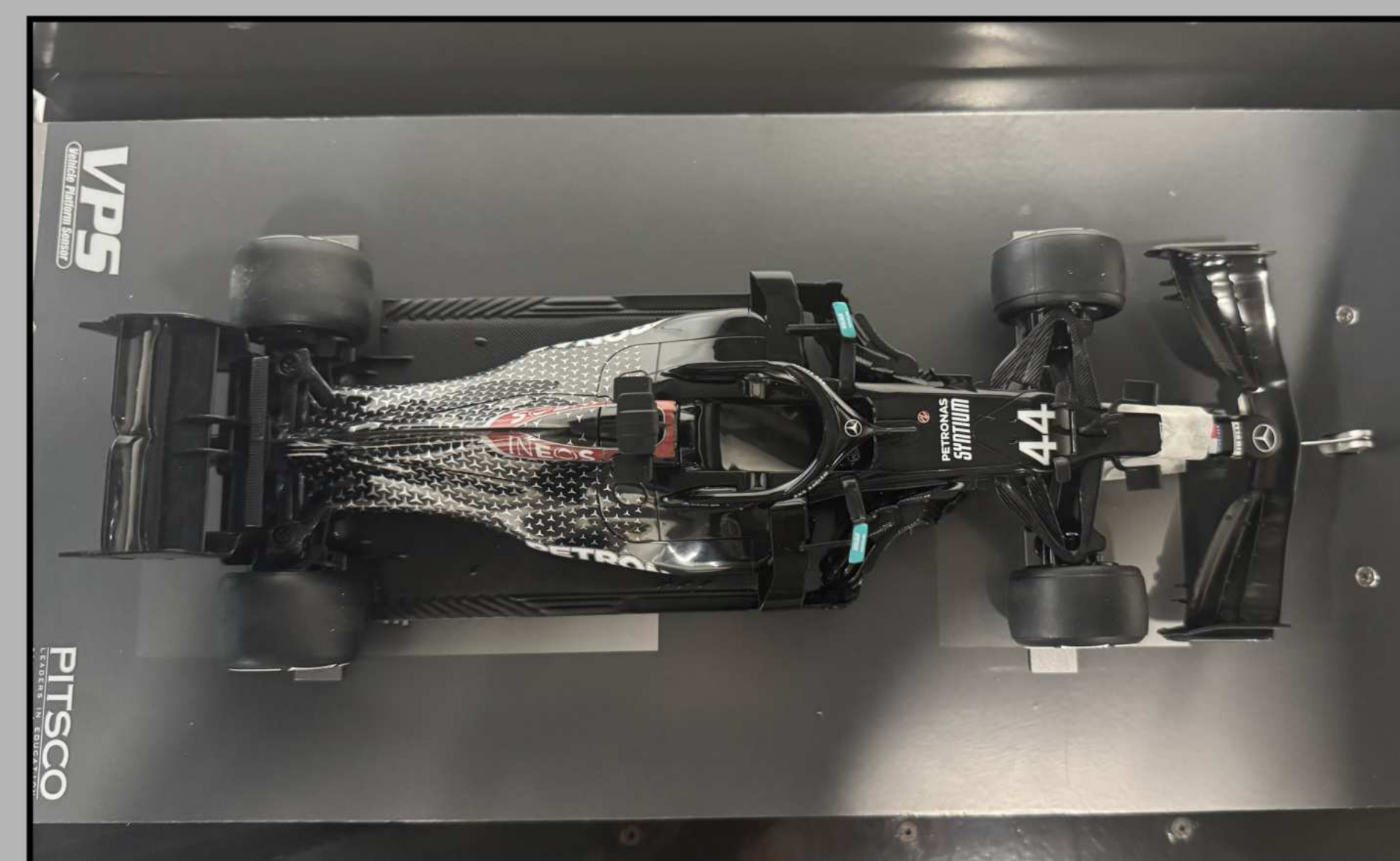


Figure 6. **Reference** remote control car inside wind tunnel, attached to force sensors

Discussion

According to the data collected from the Pitsco wind tunnel, both car models generated **negative total downforce**, also known as lift. Once this was identified, no further front wing variations were created. In an attempt to fix potential error, the weights of the cars were increased by attaching washers to the underside.



Figure 8. **Washers** attached to bottom of car

However, the data was not significantly different and both cars continued to generate lift. Therefore, the results of this experiment were **inconclusive**. It is likely that the oversimplified rear geometry of the car caused this error, as both front downforce readings are positive values.

To fix this error and continue research, the car model should be less simplified and more akin to a real 2026 regulation F1 car. Furthermore, a higher percent infill should be used in the 3D printed cars to increase weight for more realistic values.

Data

Figure 7. Data table for **10° AoA (top)** and **15° AoA (bottom) front wings** collected from Pitsco Wind Tunnel force sensors

Feet per Minute	Miles per Hour	Drag (N)	F Downforce (N)	R Downforce (N)	Total Downforce (N)	Lift:Drag	Front Lift : Drag
1000	11.36363636	1	0	-2	-2	2	0
1500	17.04545455	2	0	-4	-4	2	0
2000	22.72727273	3	1	-7	-6	2	-0.3333333333
2500	28.40909091	6	2	-9	-7	1.166666667	-0.3333333333
3000	34.09090909	14	2	-13	-11	0.7857142857	-0.1428571429
3500	39.77272727	22	4	-18	-14	0.6363636364	-0.1818181818

Feet per Minute	Miles per Hour	Drag (N)	F Downforce (N)	R Downforce (N)	Total Downforce (N)	Lift:Drag	Front Lift : Drag
1000	11.36363636	5	0	-1	-1	0.2	0
1500	17.04545455	9	0	-4	-4	0.4444444444	0
2000	22.72727273	10	0	-7	-7	0.7	0
2500	28.40909091	17	0	-10	-10	0.5882352941	0
3000	34.09090909	32	2	-15	-13	0.40625	-0.0625
3500	39.77272727	44	3	-20	-17	0.3863636364	-0.06818181818

References

Akanni, S.-D. (2002). Akanni thesis 2002_Redacted.pdf - City Research Online. City Research Online. https://openaccess.city.ac.uk/id/eprint/30896/1/Akanni%20thesis%202002_Redacted.pdf
Kachare, S. C. (2017, December). A CFD study of a multi-element front wing for a Formula one ... Scholarworks. <https://scholarworks.gvsu.edu/cgi/viewcontent.cgi?article=1864&context=theses>